

Yellow RAT

CyberDefenders Blue Team Lab -- Threat Intelligence

Category: Threat Intel | Difficulty: Easy | Tools: VirusTotal, Hybrid Analysis

The Scenario (in simple words)

During a routine security check at GlobalTech Industries, IT staff noticed strange network traffic from several employee workstations. Digging deeper, they found that employees' search engine results were being silently redirected to unfamiliar websites. Rather than a network capture, this time we were only given a single piece of evidence: the **SHA256 file hash** of the suspected malware. The whole investigation had to be done using public threat intelligence tools instead of packet analysis.

Background: what does it mean to investigate "just a hash"?

A file hash is a short fixed-length fingerprint that uniquely identifies a specific file's exact contents -- even a single byte of difference produces a completely different hash. Security researchers around the world regularly upload malware samples they encounter to shared platforms like **VirusTotal**, which scans each file with dozens of antivirus engines and lets analysts search by hash to instantly pull up everything already known about that exact file: what antivirus vendors call it, when it was first seen, what other filenames it has used, and comments from other researchers who have analyzed it before.

What Actually Happened (the story)

The provided hash belonged to a malicious .NET DLL file publicly tracked by researchers under several different names -- most commonly **Yellow Cockatoo RAT**, but also known as Jupyter Infostealer, Solarmarker, and Polazert. This is common in the security industry: different vendors and researchers who discover the same malware independently often give it different names before the community settles on shared terminology.

According to a well-known public analysis by the security firm Red Canary, once running on an infected machine this malware drops a file called **solarmarker.dat** into the victim's AppData folder, which it uses as a unique identifier for that specific infected host. It then connects out to its command-and-control (C2) server at **gogohid.com**, sending back information about the infected machine and receiving further commands -- including the browser-hijacking behavior that redirects victims' search results, matching exactly what GlobalTech's IT team observed.

Step-by-Step Investigation

Step 1 -- Search the hash on VirusTotal. Pasted the SHA256 hash from the provided hash.txt file into VirusTotal's search bar. The Detection tab immediately showed 56 out of 70 antivirus engines flagging the file as malicious, along with an automated "popular threat label" (trojan.msil/polazert) that hinted at the malware family without giving the commonly used name.

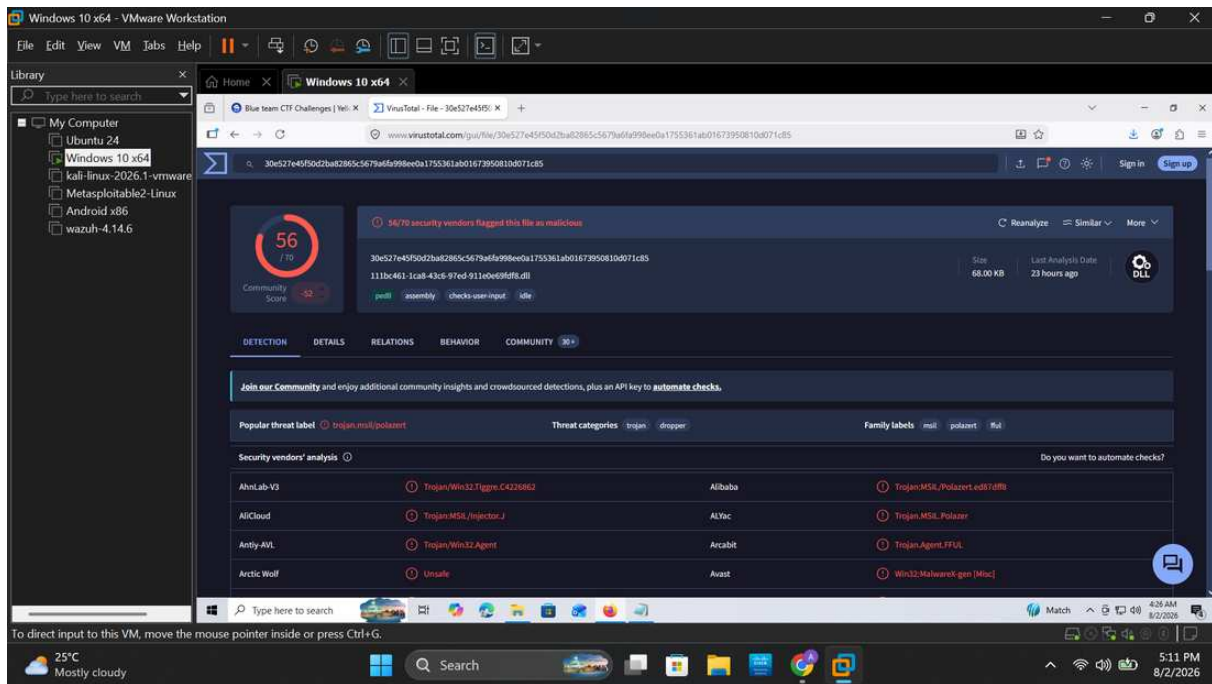


Fig 1 -- VirusTotal's Detection tab: 56/70 vendors flag the file malicious, with an automated label hinting at the Polazert family alias.

Step 2 -- Check the Community tab for analyst context. The automated label alone wasn't specific enough to match the expected answer format, so the Community tab (where human researchers leave notes, tags, and links to their own write-ups) was checked next. Comments there tagged the file with #jupyter and listed it under the "jupyter" malware family in a separate analysis report.

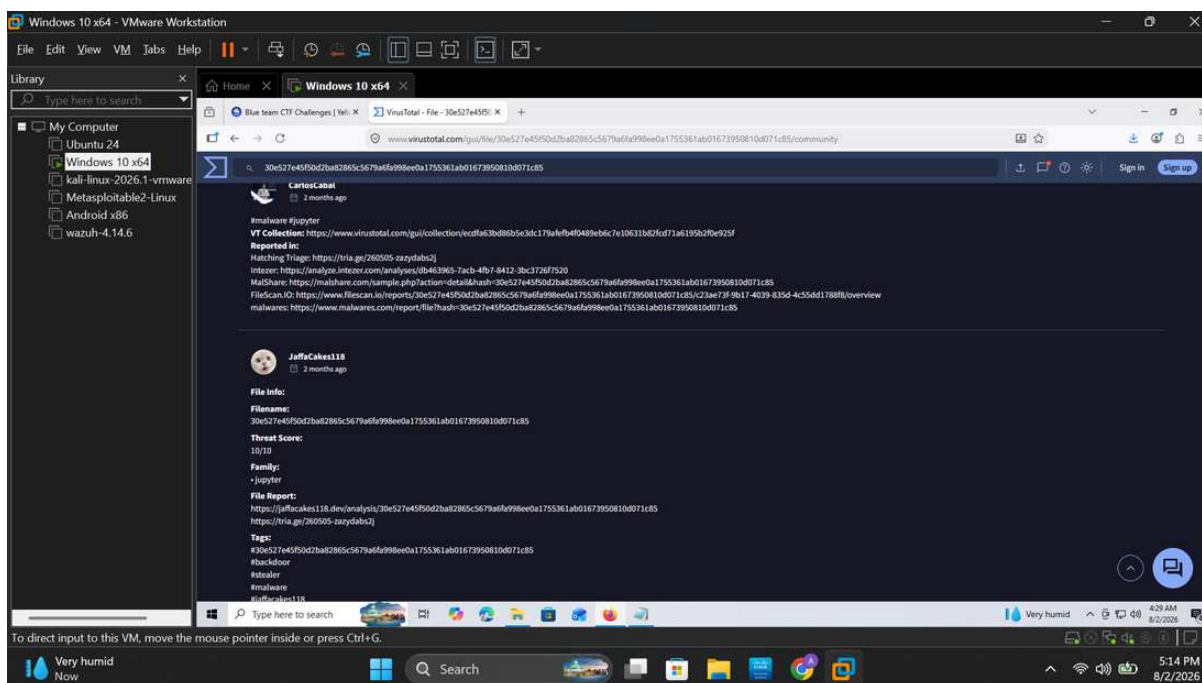


Fig 2 -- Community comments tag the sample as part of the Jupyter malware family, one of several aliases for the same threat.

Step 3 -- Pull technical metadata from the Details tab. The Details tab provided the file's compilation timestamp, its first submission date to VirusTotal, and -- critically -- a full list of every filename this exact file has been seen under across all its submissions, including a GUID-style filename used as the malware's disguise.

Step 4 -- Cross-reference with a public threat report. VirusTotal doesn't track every dropped file or C2 address, so a public search led to Red Canary's detailed writeup on this malware family (tracked there as "Yellow Cockatoo"), which explicitly documents the dropped .dat filename and the C2 domain the malware communicates with.

Final Answers

Q	Question (short)	Answer
Q1	Malware family name	Yellow Cockatoo RAT
Q2	Common filename used by the malware	111bc461-1ca8-43c6-97ed-911e0e69fdf8.dll
Q3	Compilation timestamp	2020-09-24 18:26
Q4	First submitted to VirusTotal	2020-10-15 02:47
Q5	.dat file dropped in AppData	solarmarker.dat
Q6	C2 server	gogohid.com

How Each Answer Was Found

Q1 -- Yellow Cockatoo RAT	VirusTotal's own automated label (trojan.msil/polazert) didn't match the expected 3-word answer format. Checked the Community tab instead, where human analysts had tagged the sample #jupyter and listed a "jupyter" family name -- this is one of several public aliases for the same malware. A general web search for the scenario's exact wording ("search queries redirected") confirmed this malware is most widely documented under the name Yellow Cockatoo, matching the lab's own title and the required answer format exactly.
Q2 -- GUID filename	Read directly from VirusTotal's Details tab, under the "Names" section, which lists every filename this exact file hash has been submitted or observed under. One entry, 111bc461-1ca8-43c6-97ed-911e0e69fdf8.dll, matched the required GUID-style answer format exactly (also confirmed as the file's "Original Name" in its embedded version info).
Q3 -- Compilation timestamp	Read directly from VirusTotal's Details tab, under the PE Header section, labeled "Compilation Timestamp: 2020-09-24 18:26:47 UTC" -- this is metadata embedded in the file itself by the compiler that built it, not something inferred.
Q4 -- First VirusTotal submission	Read directly from VirusTotal's Details tab: "First Submission: 2020-10-15 02:47:37 UTC". Care was taken not to confuse this with the separate "First Seen In The Wild" date shown just above it, which reflects a different (later) timestamp from a different data source.
Q5 -- solarmarker.dat	Not available on VirusTotal itself -- this required searching for and reading a public third-party analysis. A web search for the malware family plus "AppData .dat file" led to Red Canary's published threat report, which explicitly states the malware writes a randomly generated identifier string to %USERPROFILE%\AppData\Roaming\solarmarker.dat.
Q6 -- gogohid.com	Found in the same Red Canary report referenced for Q5, which documents the malware connecting to https://gogohid[.]com/gate?q=... as its command-and-control server to register the infected host and retrieve further instructions.